

RetailPro

Retail Sales & Business Analytics Dashboard

End-to-End Microsoft Power BI Portfolio Project

Tools: Power BI • Power Query • DAX • Excel • Data Modeling

Project Documentation

Business objectives, dataset, data model, DAX, dashboard pages, advanced Power BI features, insights, recommendations, and technical skills.

1. Project Overview

RetailPro is an end-to-end Power BI retail analytics project designed to provide management with a comprehensive view of sales performance, profitability, customers, products, stores, returns, and target achievement.

The project transforms raw retail transaction data into an interactive business intelligence solution using Power Query, DAX, data modeling, time intelligence, and advanced Power BI features.

Project Objective

- Monitor sales and profitability performance
- Analyze monthly, quarterly, yearly, and YTD trends
- Understand customer behavior and repeat purchasing
- Identify high-performing products and categories
- Compare store and geographical performance
- Analyze product and sales returns
- Track actual sales against targets
- Perform what-if pricing analysis
- Drill into detailed business dimensions
- Support data-driven business decisions

Tools & Technologies

- Microsoft Power BI
- Power Query
- DAX
- Microsoft Excel
- Data Modeling
- Star Schema
- Time Intelligence
- Interactive Data Visualization

2. Dataset & Data Model

The project uses a synthetic retail sales dataset created for business analytics and Power BI dashboard development. The dataset contains sales transactions, product information, customer information, store details, employee information, returns, and sales targets.

Table	Description
DimDate	Date, year, quarter, month and calendar information
DimProduct	Product, category and subcategory information
DimCustomer	Customer demographics and customer segments
DimStore	Store, region, state and city information
DimEmployee	Employee information
FactSales	Retail sales transaction data
FactReturns	Product return transactions
FactTargets	Monthly sales targets by store

Data Model

The Power BI solution follows a Star Schema architecture. The central fact table is FactSales, connected to dimension tables for filtering, grouping, and analysis.

Fact Tables

- FactSales — Primary sales transaction data
- FactReturns — Product and sales return information
- FactTargets — Monthly store-level sales targets

Dimension Tables

- DimDate — Date and time intelligence
- DimProduct — Product hierarchy and product attributes
- DimCustomer — Customer attributes and segmentation
- DimStore — Store and geographical information
- DimEmployee — Employee information
- DimMonth — Monthly bridge used for target/date analysis

Hierarchies

- Geography Hierarchy: Region → State → City
- Product Hierarchy: Product Category → Product Subcategory → Product Name

The model was designed to support efficient filtering, aggregation, time-intelligence calculations, drill-through analysis, and interactive dashboard reporting.

3. Power Query & Data Preparation

Power Query was used in Power BI to connect to the source Excel workbook, inspect the available tables, prepare the data, and ensure that the dataset was ready for analysis.

- Connecting Power BI to the Excel source workbook
- Loading the required fact and dimension tables
- Reviewing table structures and column data types
- Validating date, numeric, text, and key columns
- Ensuring sales, cost, quantity, discount, return, and target fields were available for analysis
- Preparing date-related fields required for time intelligence
- Maintaining consistent key columns for relationships
- Checking the data model for missing or inconsistent relationships
- Preparing the dataset for DAX calculations and interactive reporting

Power Query was used as the data preparation layer, while DAX was used for business calculations and analytical measures.

4. DAX & Business Calculations

DAX was used extensively to create reusable, filter-aware measures for sales, profitability, customers, products, stores, returns, targets, pricing, and time-based analysis.

Core Sales & Profitability

- Total Sales
- Total Cost
- Gross Profit
- Profit Margin %

- Total Quantity
- Total Orders
- Average Order Value

Time Intelligence

- Previous Year Sales
- YoY Sales
- YoY Sales %
- YTD Sales
- MTD Sales
- QTD Sales
- Rolling 3M Sales
- Rolling 6M Sales
- Rolling 12M Sales
- Previous Month Sales
- MoM Sales Growth %
- Previous YTD Sales
- YTD Sales Growth %

Customer Analytics

- Total Customers
- Customer Sales
- New Customers
- Repeat Customers
- Repeat Customer Rate %
- Average Customer Value
- Customer Contribution %
- Customer Rank
- Customer Profit Margin %
- Sales per Customer
- Profit per Customer
- Orders per Customer
- Average Units per Customer

Product Analytics

- Product Sales
- Product Cost
- Product Profit
- Product Margin %
- Product Quantity
- Product Rank
- Product Contribution %
- Product Profit % of Total

- Product Profit per Unit
- Product Sales % of Total
- Cumulative Product Sales
- Cumulative Product Sales %
- Products to 80% Sales
- Pareto 80% Line

Store & Geographic Analytics

- Total Stores
- Average Sales per Store
- Store Contribution %
- Store Rank
- Geographic Contribution %
- Region Rank
- State Rank
- City Rank

Returns & Net Performance

- Total Returned Quantity
- Total Returned Sales
- Return Rate %
- Returned Sales
- Sales Return Rate %
- Net Sales
- Net Profit
- Net Margin %
- Return Amount % of Sales
- Net Quantity
- Returned Orders
- Return Order Rate %
- YoY Net Sales %
- YTD Net Sales
- YoY Net Profit %

Target Analysis

- Total Sales Target
- Target Variance
- Target Achievement %
- Target Variance %
- Target Status
- Target Performance Indicator

Discount & Pricing Analysis

- Total Discount

- Average Discount %
- Discounted Sales
- Full-Price Sales
- Average Selling Price
- Discount Impact %
- Discounted Sales Margin %

Advanced Sales Performance

- Units per Order
- Profit per Order
- Profit per Customer
- Profit per Unit
- Sales per Customer
- Orders per Customer
- Average Units per Customer

What-If Analysis

A Price Change % what-if parameter was implemented to simulate changes in selling prices. Scenario measures include Scenario Sales, Scenario Profit, and Scenario Profit Margin %.

Dynamic Reporting & Status Measures

- Selected Year
- Selected Month
- Selected Store
- Dynamic Report Title
- Dashboard Title
- Dashboard Context
- Sales Performance Status
- Profit Performance Status
- Return Risk Status
- YoY Sales Indicator
- Target Performance Indicator

DAX Techniques Used

- CALCULATE
- SUM
- SUMX
- DIVIDE
- DISTINCTCOUNT
- FILTER
- RANKX
- ALL
- ALLSELECTED
- REMOVEFILTERS

- SELECTEDVALUE
- HASONEVALUE
- SWITCH
- SAMEPERIODLASTYEAR
- DATEADD
- TOTALYTD
- TOTALMTD
- TOTALQTD
- DATESINPERIOD
- TREATAS
- VAR and RETURN

5. Dashboard Pages & Visualizations

Executive Dashboard

KPIs: Total Sales, Gross Profit, Profit Margin %, Total Orders, Total Customers, Net Sales, Return Rate %, Repeat Customer Rate %, Target Achievement %. Visuals: Sales Trend, Profit Trend, Sales by Category, Actual Sales vs Target, Sales by Region, Top 10 Products by Sales.

Sales Analytics

KPIs: Total Sales, Rolling 6M Sales, Rolling 3M Sales, YoY Sales Growth %, Total Orders, Average Order Value, Total Quantity. Visuals: Monthly Sales & YoY Growth, Monthly Sales vs Rolling 3M Average, Top 10 Stores by Sales, Sales by Customer Segment, Sales by Geography, Annual Sales vs Gross Profit.

Customer Analytics

KPIs: Total Customers, New Customers, Repeat Customers, Repeat Customer Rate %, Average Customer Value. Visuals: Customers & Sales by Segment, Customer Sales by Region, Customer Segment Sales Ranking by Year, Customer Sales Decomposition, Customer Sales by Segment & Region.

Product Analytics

KPIs: Product Sales, Product Profit, Product Margin %, Product Quantity, Product Profit per Unit. Visuals: Product Sales Pareto Analysis, Product Profitability Matrix, Product Sales Mix by Category, Product Category Sales Ranking by Year.

Store Analytics

KPIs: Total Stores, Total Sales, Average Sales per Store, Total Orders, Profit Margin %. Visuals: Store Performance Matrix, Store Sales Contribution by Region, Store Sales vs Profitability, Store Sales by City.

Returns Analytics

KPIs: Total Returned Sales, Total Returned Quantity, Return Rate %, Sales Return Rate %, Net Margin %. Visuals: Return Sales Trend, Return Sales by Reason, Return Reason Ranking by Year, Return Sales Decomposition, Top 10 Stores by Return Rate %, Return Quantity by Product Category.

Target Analytics

KPIs: Total Sales, Sales Target, Target Achievement %, Target Variance, Target Variance %. Visuals: Monthly Sales vs Target Achievement, Target Variance Contribution by Region, Store Sales vs Target, Store Target Performance Matrix.

Interactive Power BI Features

- Synchronized slicers
- Year, Month, Region, Product Category and Store filtering
- Reset Filters button

- Page navigation
- Product, Customer and Store drill-through
- Field Parameter for dynamic metrics
- What-If Price Change analysis
- Interactive hierarchies
- Tooltips
- Cross-filtering
- Dynamic dashboard context

6. Power BI Features & Advanced Functionality

Synchronized Slicers

Slicers were synchronized across all report pages to provide consistent filtering.

Drill-Through Analysis

Drill-through functionality was implemented for Product, Customer, and Store analysis.

Hierarchies

Geography and Product hierarchies support structured drill-down.

Field Parameter

Metric Selector dynamically switches between Total Sales, Gross Profit, Total Quantity, Total Orders, and Average Order Value.

What-If Analysis

Price Change % enables interactive price scenarios from -20% to +20%.

Reset Filters

A bookmark-based Reset Filters button returns the dashboard to its default filter state.

Page Navigation

A page navigator provides navigation across the seven analytical pages.

Dynamic Dashboard Context

Dynamic measures display selected Year, Month, and Store context.

Cross-Filtering & Interactions

Selections in visuals dynamically filter or highlight related visuals.

Dashboard Design

A consistent dark and electric-cyan theme provides a professional, modern, portfolio-ready experience.

7. Key Business Insights

Sales & Profitability

Total Sales are approximately ■13.17 billion, Gross Profit approximately ■2.99 billion, Gross Profit Margin approximately 22.71%, Net Sales approximately ■12.18 billion, Net Profit approximately ■2.00 billion, and Net Margin approximately 16.45%.

Sales Performance

Total Orders are approximately 99,768, Total Quantity is approximately 450,658 units, and Average Order Value is approximately ■131.96K. Sales performance shows growth across the later years of the analysis.

Customer Insights

The dataset contains 10,000 customers. The analysis identifies a 100% repeat customer rate within the available transaction data.

Product Insights

Product performance varies across categories, subcategories, and individual products. Pareto analysis shows that approximately 257 products account for 80% of cumulative sales.

Store & Geographic Insights

Store performance varies across regions, states, cities, and individual stores. Store ranking and contribution analysis support location-level performance management.

Returns Insights

Total Returned Quantity is approximately 33,600 units. Total Returned Sales are approximately ■986.20 million. Return Rate is approximately 7.46% and Sales Return Rate is approximately 14.97%.

Target Performance

Total Sales Target is approximately ■357.19 million. Overall Target Achievement is approximately 368.57%, with Target Variance of approximately ■959.31 million and status Above Target.

Discount & Pricing

Total discount impact is approximately ■994.09 million and Average Discount Percentage is approximately 7.02%. What-If analysis demonstrates how price changes can affect sales, profit, and margin.

Overall, the business demonstrates strong sales and profitability, positive sales growth, a highly active repeat customer base, and performance above defined targets. Returns and discounting nevertheless have a meaningful effect on net sales and profitability.

8. Business Recommendations

Improve Profitability

Focus on products and stores that generate strong profit rather than evaluating performance only by sales revenue. Monitor profit margin at product, category, store, and regional levels.

Optimize Discount Strategy

Monitor discount levels and their effect on margins. Prefer targeted promotions and use the Price Change What-If analysis before major pricing changes.

Optimize Product Portfolio

Prioritize high-performing products, use Pareto analysis to focus on major revenue contributors, and review low-performing products.

Strengthen Customer Relationships

Focus on retention, identify high-value customers, develop targeted loyalty programs, and analyze customer segments and regions.

Improve Store Performance

Benchmark stores within regions, investigate high-sales/low-profit stores, replicate successful practices, and use target achievement to prioritize support.

Strengthen Geographic Strategy

Compare regions, states, and cities, identify growth areas, investigate underperforming locations, and use geographic contribution for resource allocation.

Reduce Product Returns

Investigate major return reasons, products and categories with high return activity, and stores with high return rates.

Improve Target Management

Monitor target achievement at monthly, regional, and store levels and review target assumptions regularly.

Monitor Sales Trends

Use YoY, MoM, YTD, and rolling-period metrics to identify changes in business momentum early.

Overall, the organization should adopt a balanced strategy focused on revenue, profitability, customer retention, return reduction, discount optimization, and store performance.

9. Technical Skills Demonstrated

Microsoft Power BI

- Power BI Desktop
- Interactive dashboard development
- KPI development
- Visual interactions
- Cross-filtering
- Drill-through
- Bookmarks
- Synchronized slicers
- Tooltips
- Field Parameters
- What-If Parameters
- Hierarchies
- Dynamic reporting context

Power Query

- Excel connectivity
- Data preparation and transformation
- Data type validation
- Fact and dimension preparation
- Data quality checks

DAX

- Measure creation
- Filter context
- CALCULATE
- SUM
- SUMX
- DIVIDE
- DISTINCTCOUNT

- FILTER
- RANKX
- ALL
- ALLSELECTED
- REMOVEFILTERS
- SELECTEDVALUE
- HASONEVALUE
- SWITCH
- TREATAS

Data Modeling

- Star Schema
- Fact and dimension modeling
- Relationship management
- Date dimension
- Monthly bridge table
- Hierarchies
- Filter propagation
- Model validation

Business Analytics

- Sales
- Profitability
- Customer
- Product
- Store
- Geographic
- Returns
- Target
- Discount
- Pricing scenario
- Pareto analysis

Data Visualization

- KPI Cards
- Line Charts
- Column Charts
- Bar Charts
- Combo Charts
- Matrix
- Tables
- Donut Charts
- Treemaps

- Ribbon Charts
- Scatter Charts
- Waterfall Charts
- Maps
- Decomposition Trees
- Slicers

10. Project Structure & File Organization

The project is organized into separate files and folders to keep the Power BI report, source dataset, documentation, and supporting assets structured and easy to maintain.

RetailPro_PowerBI_Project/

■■■■ RetailPro_PowerBI_Project.pbix

■■■■ RetailPro_PowerBI_Project.xlsx

■■■■ README.md

■■■■ Screenshots/

■ ■■■■ Executive_Dashboard.png

■ ■■■■ Sales_Analytics.png

■ ■■■■ Customer_Analytics.png

■ ■■■■ Product_Analytics.png

■ ■■■■ Store_Analytics.png

■ ■■■■ Returns_Analytics.png

■ ■■■■ Target_Analytics.png

■■■■ Documentation/

■■■■ RetailPro_Project_Documentation.pdf

Portfolio Sharing

The README.md file provides the main GitHub project documentation. The PBIX file demonstrates the complete interactive reporting solution, while screenshots provide visual proof when an interactive Power BI report is not available.